

NAND & NOR Gates

- NAND and NOR gates are universal gates, meaning any digital circuit can be built using only one of them.
 - In CMOS logic, NAND & NOR gates have simpler and faster transistor-level implementations than AND & OR gates because they require **fewer transistors** and avoid additional inverter stages.
 - The **cost** of creating NAND & NOR is less than that for AND & OR gates.
 - Any AND-OR network (SOP) can be implemented as a **NAND-NAND** network.
 - Any OR-AND network (POS) can be implemented as a **NOR-NOR** network.
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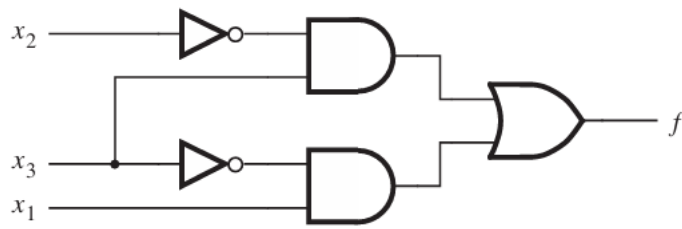
- NAND Gate

1. AND logic using NAND:

2. OR logic using NAND:

3. NOT logic using NAND

Example 1: Implement the following circuit using **only NAND** gates (i.e., NAND-NAND network).



- NOR Gate

1. AND logic using NOR:

2. OR logic using NOR:

3. NOT logic using NOR

Example 2: Implement the following circuit using **only NOR** gates (i.e., NOR-NOR network).

